



VP AIR TEST NEET (2025-26)

PART TEST - 06

DROPPER

DURATION: 180 Minutes

Date: 30-11-2025

M. MARKS: 720

Topics Covered

Physics:	Oscillations, Waves, Electric Charges and Fields
Chemistry:	Organic Chemistry - Isomerism, Organic Chemistry - General Organic Chemistry
Botany:	Plant kingdom
Zoology:	Chemical Coordination and Integration

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR Instructions:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. When the displacement of a simple harmonic oscillator is one third of its amplitude, the ratio of total energy to the kinetic energy is denoted as $\frac{x}{8}$.

The value of x is:

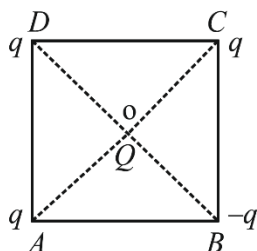
- (1) 9 (2) 8
(3) 12 (4) 10

2. The equation of a stationary wave along a stretched string is given by $y = 5 \sin\left(\frac{2\pi}{3}x\right) \cos(40\pi t)$

(where x and y are in cm and t is in s). The separation between two adjacent nodes is:

- (1) 1.5 cm
(2) 3 cm
(3) 6 cm
(4) 4 cm

3. Four-point charges are fixed at vertices of square as shown in figure. The net electric force on point positive charge Q placed at center of square is: (Distance between vertex and center of square is r)

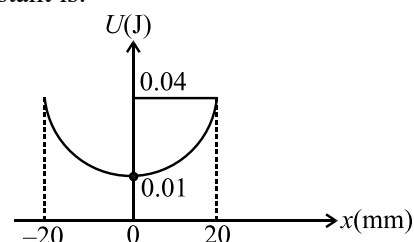


- (1) $\frac{2kqQ}{r^2}$ away from $-q$ charge
(2) $\frac{2kqQ}{r^2}$ towards $-q$ charge
(3) $\frac{kqQ}{r^2}$ away from $-q$ charge
(4) $\frac{kqQ}{r^2}$ towards $-q$ charge

4. V_H and V_D denote the speed of sound in humid air and dry air respectively at same temperature and pressure. Choose the **correct** relation.

- (1) $V_H > V_D$
(2) $V_H = V_D$
(3) $V_H < V_D$
(4) Data insufficient

5. The variation of potential energy of harmonic oscillator is as shown in figure. The value of spring constant is:



- (1) 1×10^2 N/m (2) 150 N/m
(3) 0.667×10^2 N/m (4) 3×10^2 N/m

6. Given below are two statements:

Statement I: Net charge of particle can be $2.5e$ where e is charge of electron.

Statement II: Total amount of charge in nature is always conserved.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

7. The insulation property of air breaks down at external field of magnitude $E = 3 \times 10^6$ V/m. The maximum charge that can be given to a uniform conducting sphere of diameter 5 m, placed in air without air losing its insulation property is approximately;

- (1) 2×10^{-5} C (2) 2×10^{-4} C
(3) 2×10^{-3} C (4) 3×10^{-3} C

8. The speed of sound in a medium at constant temperature

- (1) is proportional to the pressure of the medium.
(2) is proportional to the square of pressure of the medium.
(3) is proportional to the square root of pressure of the medium.
(4) does not depend on pressure of medium.

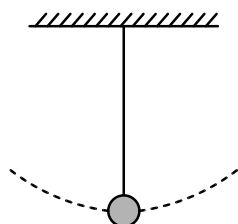
9. An electric dipole is placed at an angle of 30° with an uniform electrostatic field having intensity 2×10^5 N/C. It experiences a torque equal to 4 Nm. The positive charge on the dipole, if the dipole length is 2 cm, is;

- (1) 4 mC (2) 7 mC
(3) 8 mC (4) 2 mC

10. The first overtone of an open organ pipe has the same frequency as the first overtone of a closed pipe of length L . The length of the open organ pipe will be:

- (1) L (2) $\frac{2L}{3}$
 (3) $\frac{3L}{4}$ (4) $\frac{4L}{3}$

11. The figure shows a pendulum at the bottom of its swing.



Which is the direction of the net force acting on the pendulum bob at this instant?

- (1) $\uparrow$
 (2) $\downarrow$
 (3) $\searrow$
 (4) The net force is zero
12. An infinite number of charges, each of charge $1 \mu\text{C}$, are placed on the x -axis with co-ordinates $x = 1\text{m}, 2\text{m}, 4\text{m}, 8\text{m}, \dots, \infty$. If a charge 1 C is kept at the origin, then net force acting on 1 C charge:
- (1) 9000 N (2) 12000 N
 (3) 24000 N (4) 36000 N
13. If point charge q is placed on centre of cubical gaussian surface then electric flux due to it through one face of cube is;
- (1) $\frac{q}{\epsilon_0}$ (2) $\frac{q}{3\epsilon_0}$
 (3) $\frac{q}{6\epsilon_0}$ (4) $\frac{q}{8\epsilon_0}$
14. If the vertical spring-mass system executing SHM is completely dipped in a non-viscous liquid, then:
- (1) the mean position changes, but time period remain same.
 (2) the time period changes, but mean position remain same.
 (3) Both the time period and mean position change.
 (4) Both the time period and mean position remain the same.

15. A particle of mass 100 g , executing SHM passes the mean position with speed 100 m/s . If angular frequency of SHM is $10\pi \text{ rad/s}$ then amplitude of SHM is:

- (1) $\left(\frac{10}{\pi}\right) \text{ m}$
 (2) $(10\pi) \text{ m}$
 (3) $\left(\frac{5}{\pi}\right) \text{ m}$
 (4) $(100\pi) \text{ m}$

16. Match the laws given in **List-I** to their relevant description in **List-II**.

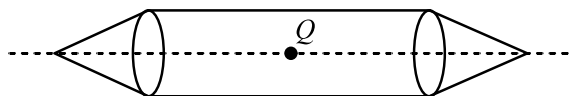
List-I		List-II	
A.	Coulomb's law	I.	Total electric flux through a gaussian surface is proportional to charge enclosed in the gaussian surface.
B.	Gauss's law	II.	Vector sum of forces.
C.	Principle of superposition	III.	Force is inversely proportional to square of distance
D.	Quantization of charge	IV.	Charge exists in discrete packets

- (1) A-II, B-III, C-I, D-IV
 (2) A-III, B-I, C-II, D-IV
 (3) A-I, B-IV, C-III, D-II
 (4) A-I, B-II, C-III, D-IV
17. If the displacement x (in meter) and velocity v (in m/s) of a particle executing simple harmonic motion are related by the expression $4v^2 = 25 - x^2$, then its time period is:
- (1) π second
 (2) 2π second
 (3) 4π second
 (4) 6π second
18. consider uniformly charged non-conducting solid sphere of radius r . Magnitude of electric field (E) due to it at distance x from its centre is; ($x < r$)
- (1) $E \propto \frac{1}{x^2}$
 (2) $E \propto \frac{1}{x^3}$
 (3) $E \propto x$
 (4) $E \propto \frac{1}{x}$

19. A closed and open organ pipe of same length produce 2 beats when they are set into vibrations simultaneously in their fundamental mode. The length of the open organ pipe is now halved and that of the closed pipe is doubled. The number of beats produced will be (while vibrating in fundamental mode):

(1) 7 (2) 8
(3) 5 (4) 9

20. The two ends of a hollow cylinder are covered with hollow cones as shown in the figure. A charge Q is placed exactly at the center of cylinder on its axis. What is the electric flux associated with each cone, if flux through curved part of hollow cylinder is ϕ_0 .



(1) $\frac{Q}{2\epsilon_0}$ (2) $\frac{1}{2}\left(\frac{Q}{\epsilon_0} - \phi_0\right)$
(3) $\frac{Q}{3\epsilon_0}$ (4) $\frac{Q}{\epsilon_0} - \phi_0$

21. Force between two point charges Q_1 and Q_2 separated by distance r is 25 N. It can be reduced to 5 N if the separation between them is made:

(1) $\frac{r}{\sqrt{5}}$
(2) $\frac{r}{2}$
(3) $\sqrt{5}r$
(4) $2r$

22. Two particles execute S.H.M. along the same line at the same frequency. They move in opposite direction at the mean position. The phase difference at that instant will be:

(1) 2π radian
(2) $\frac{2\pi}{3}$ radian
(3) π radian
(4) $\frac{\pi}{2}$ radian

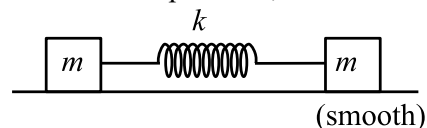
23. Two pendulum begin to swing simultaneously. The first pendulum makes 9 full oscillations in the time when the second makes 7. The ratio of length of the first and second pendulum is:

(1) $\frac{9}{7}$ (2) $\frac{7}{9}$
(3) $\frac{49}{81}$ (4) $\frac{81}{49}$

24. The net charge given to an isolated conducting solid sphere:

(1) must be distributed uniformly on the surface.
(2) may be distributed uniformly on the surface.
(3) must be distributed uniformly in the volume.
(4) may be distributed uniformly in the volume.

25. Consider the situation as shown in figure. If the blocks are pressed slightly towards each other and released, they will execute simple harmonic motion. The time period is;



(1) $2\pi\sqrt{\frac{m}{k}}$ (2) $2\pi\sqrt{\frac{m}{2k}}$
(3) $2\pi\sqrt{\frac{2m}{k}}$ (4) $2\pi\sqrt{\frac{m}{9k}}$

26. Torque acting on electric dipole of dipole moment $\vec{P}$ when placed in uniform electric field $\vec{E}$ is;

(1) $\vec{E} \times \vec{P}$
(2) $\vec{P} \times \vec{E}$
(3) $\vec{E} \cdot \vec{P}$
(4) Zero

27. Given below are two statements:

Statement I: In transverse mechanical waves, the constituents of the medium oscillates perpendicular to the direction of wave propagation.

Statement II: In longitudinal waves, the constituents of the medium oscillates along the direction of wave propagation.

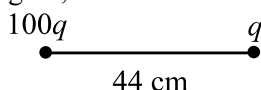
In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

28. A sitar wire is replaced by another wire of the same length and material with double the radius. If the tension in the wire remains the same, the fundamental frequency of new wire will become:

(1) nine times
(2) double
(3) half
(4) remains same

29. At what distance from the point charge $100q$ as shown in figure, the electric field is zero?



- (1) 4 cm
(2) 11 cm
(3) 22 cm
(4) 40 cm
30. What is the phase difference between two particles executing SHM along the x axis with equations $x_1 = A \sin(\omega t)$ and $x_2 = A \cos\left(\omega t + \frac{\pi}{4}\right)$

- (1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$
(3) $\frac{3\pi}{4}$ (4) $\frac{2\pi}{3}$

31. An electric dipole is placed inside a non-conducting uniformly charged spherical shell. Choose the **correct** option.

- (1) Both, net force and net torque on the dipole is non-zero.
(2) Net force on the dipole is zero but net torque is non-zero.
(3) Net force on the dipole is non-zero but net torque is zero.
(4) Both net force and net torque on the dipole is zero.

32. In a guitar, two strings A and B , made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly increased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be:

- (1) 524 Hz
(2) 536 Hz
(3) 537 Hz
(4) 523 Hz

33. Two similar, very small conducting spheres having charges $40 \mu\text{C}$ and $-20 \mu\text{C}$ are some distance apart. Now they are touched and kept at the same distance. The ratio of the initial to the final force between them is:

- (1) 8 : 1 (2) 4 : 1
(3) 1 : 8 (4) 1 : 1

34. A transverse wave is represented by the equation $y = y_0 \sin \frac{2\pi}{\lambda}(vt - x)$. For what value of λ is the maximum particle velocity equal to three times the wave velocity?

- (1) $\lambda = 2\pi y_0$ (2) $\lambda = \frac{2\pi y_0}{3}$
(3) $\lambda = \frac{\pi y_0}{2}$ (4) $\lambda = \pi y_0$

35. The particle executing simple harmonic motion has a kinetic energy $K_0 \cos^2 \omega t$. The maximum values of the potential energy and the total energy are respectively.

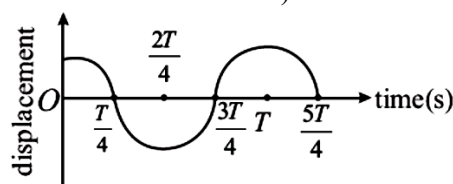
- (1) K_0 and K_0
(2) 0 and $2K_0$
(3) $\frac{K_0}{2}$ and K_0
(4) K_0 and $2K_0$

36. Three harmonic waves having equal frequency ν and same intensity I_0 , have phase angles 0 , $\frac{\pi}{4}$ and

$-\frac{\pi}{4}$ respectively. When they are superimposed the intensity of the resultant wave is close to:

- (1) $5.8 I_0$
(2) $3 I_0$
(3) I_0
(4) $0.2 I_0$

37. The displacement time graph of a particle executing S.H.M is given in figure (sketch is schematic and not to scale)



Which of the following statements is/are true for this motion?

- A. The force is zero at $t = \frac{3T}{4}$
B. The acceleration is maximum at $t = T$
C. The speed is maximum at $t = \frac{T}{4}$
D. The P.E. is equal to K.E. of the oscillation at $t = \frac{T}{2}$

- (1) B, C and D
(2) A and D
(3) A, B and C
(4) A, B and D

38. A spring mass system has time period T . If the spring is cut into four identical parts and same mass is loaded to one of them, then new time period of vibration is:

- (1) T (2) $\frac{T}{2}$
(3) $\frac{T}{4}$ (4) $2T$

39. A wave is represented by the equation

$$y = A \sin\left(10\pi x + 15\pi t + \frac{\pi}{6}\right)$$

where x is in metre and t in second. The expression represents.

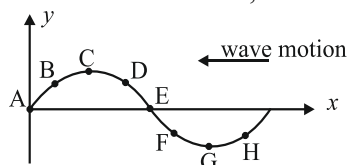
- (1) A wave travelling in negative x -direction with a velocity of 1.5 m/s
(2) A wave travelling in positive x -direction with a velocity of 1.5 m/s
(3) A wave traveling in positive x -direction with wavelength 0.2 m
(4) A wave travelling in negative x -direction with a velocity of 150 m/s
40. Three point charges, each of $+q$, are placed at the vertices of an equilateral triangle. The charge needed at the in centre of the triangle for all the charges to be in equilibrium is;

- (1) $-\frac{q}{\sqrt{3}}$ (2) $+\frac{q}{\sqrt{3}}$
(3) $+\sqrt{3}q$ (4) $-\sqrt{3}q$

41. A simple pendulum is displaced from its mean position to initiate oscillations. The torque of which of the following force acting on the pendulum tries to restore it back to the mean position?

- (1) Tension in the string
(2) Centripetal force
(3) Centrifugal force
(4) Component of weight

42. A transverse wave is travelling along a stretched string from right to left. The figure shown, represents the shape of the string (snapshot) at a given instant. At this instant;



- (1) the particles at A, B and H have downward velocity
(2) the particles at D, E and F have downward velocity
(3) the particles at C, E and G have zero velocity
(4) the particles at A and F have maximum velocity

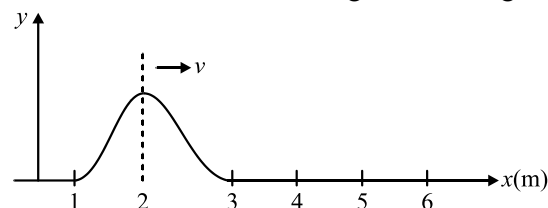
43. Match the resonance condition in **List-I** with the corresponding fundamental frequency given in **List-II**. (v is speed of sound wave in pipe)

List-I (Resonance Condition)		List-II (Frequency Formula)	
(A)	Fundamental frequency of Open Pipe	(I)	$\frac{v}{L}$
(B)	Second Harmonic of Open Pipe	(II)	$\frac{v}{4L}$
(C)	First Overtone of Closed Pipe	(III)	$\frac{v}{2L}$
(D)	Fundamental frequency of Closed Pipe	(IV)	$\frac{3v}{4L}$

Choose the **correct** answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-II, B-I, C-IV, D-III
(4) A-I, B-II, C-III, D-IV

44. Wave pulse on a string shown in figure is moving to the right without changing shape. Consider two particles at positions $x_1 = 1.5$ m and $x_2 = 2.5$ m. Transverse velocities of particle at positions x_1 , and x_2 at the moment shown in figure are along.

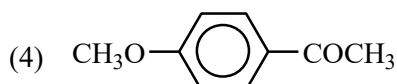
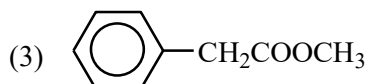
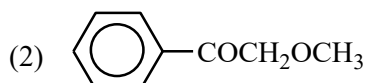
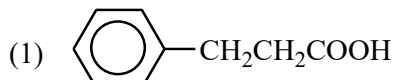
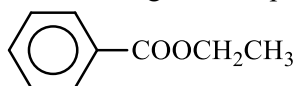


- (1) positive y -axis and positive y -axis respectively
(2) negative y -axis and positive y -axis respectively
(3) positive y -axis and negative y -axis respectively
(4) negative y -axis and negative y -axis respectively

45. A wire of length L and mass per unit length $6.0 \times 10^{-3} \text{ kg m}^{-1}$ is put under tension of 540 N. Two consecutive frequencies that it resonates at are 420 Hz and 490 Hz. Then L in meters is:

- (1) 1.1 m (2) 5.1 m
(3) 2.1 m (4) 8.1 m

46. Metamer of given compound is:



47. Which among the following groups show +R effect?

- (1) $-\text{CN}$
(2) $-\text{CHO}$
(3) $-\text{OCOCH}_3$
(4) $-\text{NO}_2$

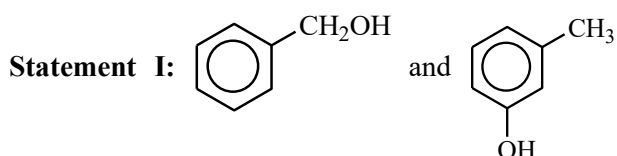
48. Tautomerism is exhibited by:

- (1) PhCOPh (2) PhCOCH_3
(3) $(\text{CH}_3)_3\text{C}-\text{CHO}$ (4) HCHO

49. Which of the following species is **not** an electrophile?

- (1) NH_4^+ (2) Cl^+
(3) B_2H_6 (4) AlCl_3

50. Given below are two statements:



are chain isomers.

Statement II: n-butyl alcohol and sec-butyl alcohol are chain isomers of each other.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

51. Most stable carbocation among the following is:

- (1) PhCH_2^+ (2) CH_3CH_2^+
(3) $\text{PhC}^+(\text{CH}_3)_2$ (4) $(\text{CH}_3)_2\text{CH}^+$

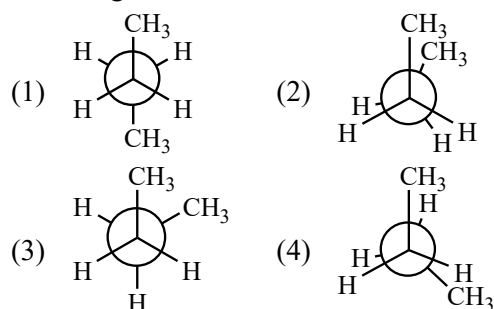
52. Maximum number of structural isomers possible on monochlorination of toluene is:

- (1) 3 (2) 4
(3) 8 (4) 6

53. Strongest acid among the following is:

- (1) CH_3COOH (2) CHCl_2COOH
(3) CCl_3COOH (4) CH_2ClCOOH

54. Most stable conformer of butane among the following is:



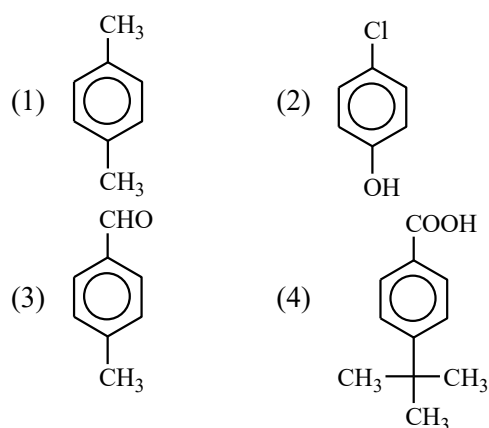
55. Temporary effect among the following is:

- (1) Inductive effect
(2) Hyperconjugation
(3) Resonance
(4) Electromeric effect

56. Which among the following compounds does **not** exhibit geometrical isomerism?

- (1) But-2-ene (2) Pent-2-ene
(3) Hex-3-ene (4) Isobutylene

57. In which of the following compounds, inductive, mesomeric and hyperconjugation effects are observed?



58. How many stereoisomers does the molecule $\text{CH}_2=\text{CH}-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2\text{CH}_3$ have?

- (1) 2 (2) 4
(3) 6 (4) 8

59. The **correct** match is:

- (1)  : aromatic
 (2)  : anti-aromatic
 (3)  : non-aromatic
 (4)  : anti-aromatic

60. Consider the following statements regarding cis-but-2-ene and trans-but-2-ene

- (A) Cis-isomer has more dipole moment than trans-isomer.
 (B) Both are diastereomers of each other.
 (C) Both are interconvertible at room temperature.
 The **correct** statements are:

- (1) A and B only (2) B and C only
 (3) A and C only (4) A, B and C

61. Most stable carbanion among the following is:

- (1) $\bar{\text{C}}\text{H}_3$ (2) $\text{CH}_3-\bar{\text{C}}\text{H}_2$
 (3) $\text{HC}\equiv\text{C}^-$ (4) $\text{CH}_2=\bar{\text{C}}\text{H}$

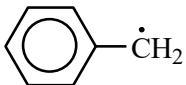
62. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Total number of conformers of ethane are infinite.

Reason R: There is almost free rotation about C–C bond in ethane.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true but R is NOT the correct explanation of A.

63. Most stable free radical among the following is:

- (1) $\text{CH}_2=\text{CH}-\dot{\text{C}}\text{H}_2$
 (2)  $\dot{\text{C}}\text{H}_2$
 (3) $\text{CH}_3-\dot{\text{C}}\text{H}_2$
 (4) $\text{CH}_2=\dot{\text{C}}\text{H}$

64. Number of position isomers of dichlorobenzene is:

- (1) 2 (2) 3
 (3) 4 (4) 5

65. Given below are two statements:

Statement I: CH_3^+ is a trigonal planar species.

Statement II: CH_3^+ behave as an electrophile.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are incorrect.

66. Number of geometrical isomers possible for $\text{C}_2\text{H}_5-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}_3$ is:

- (1) 4 (2) 2
 (3) 6 (4) 8

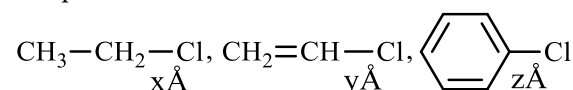
67. Electromeric effect is observed in:

- (1) $\text{CH}_3\text{CH}_2\text{OH}$ (2) CH_3OCH_3
 (3) CH_3COCH_3 (4) $\text{CH}_3\text{CH}_2\text{Cl}$

68. Most acidic hydrogen is present in:

- (1) $\text{C}_2\text{H}_5\text{OH}$ (2) CH_3COCH_3
 (3) H_2O (4) C_6H_6

69. **Correct** order of C–Cl bond length in given compounds is:



- (1) $x > y > z$ (2) $z > y > x$
 (3) $y > z > x$ (4) $z > x > y$

70. Number of structure isomers corresponding to molecular formula $\text{C}_3\text{H}_9\text{N}$ is:

- (1) 2 (2) 3
 (3) 4 (4) 5

71. Given below are two statements:

Statement I: Hyperconjugation involves overlapping of σ -p orbitals.

Statement II: Hyperconjugation is also known as no bond resonance.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Both Statement I and Statement II are correct.
 (2) Statement I is correct but Statement II is incorrect.
 (3) Statement I is incorrect but Statement II is correct.
 (4) Both Statement I and Statement II are incorrect.

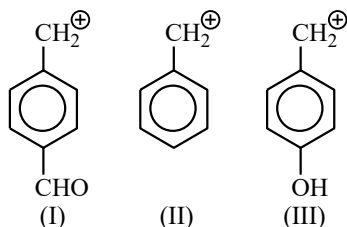
72. The compound C_3H_6O may be a:

- I. Carbonyl compound
- II. Cyclic alcohol
- III. Unsaturated ether.

Choose the **correct** options:

- (1) I and II only
- (2) II and III only
- (3) I and III only
- (4) I, II and III

73. The **correct** order of stability of given carbocations is:



- (1) III > II > I
- (2) I > II > III
- (3) III > I > II
- (4) II > III > I

74. Match the **List-I** with **List-II**.

List-I		List-II	
(A)	n-butane and iso-butane	(I)	Tautomers
(B)	Isobutyl alcohol and tert-butyl alcohol	(II)	Chain isomers
(C)	Propanamide and hydroxy propanimine	(III)	Functional isomers
(D)	Butanone and butanal	(IV)	Position isomers

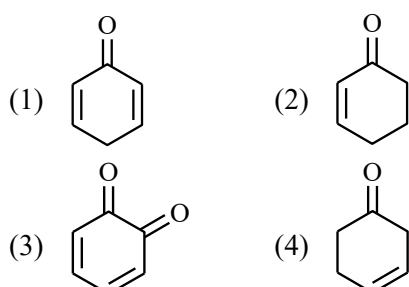
Choose the **correct** answer from the options given below;

- (1) A-IV; B-III, C-II; D-I
- (2) A-II; B-IV, C-I; D-III
- (3) A-I; B-IV, C-II; D-III
- (4) A-II; B-I, C-IV; D-III

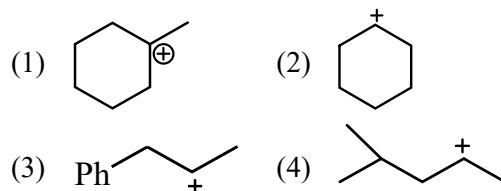
75. Which among the following does **not** behave as nucleophile?

- (1) C_6H_6
- (2) C_2H_4
- (3) C_2H_2
- (4) C_2H_6

76. Maximum enol content is observed in:



77. In which of the following carbocations rearrangement take place?

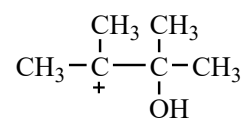


78. Number of alkenes (including stereoisomers) possible for hydrocarbon C_4H_8 is:

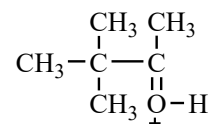
- (1) 4
- (2) 3
- (3) 6
- (4) 5

79. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: $CH_3-C(CH_3)_2-C(CH_3)_2-OH$ is more stable than $CH_3-C(CH_3)_2-C(CH_3)_2^+$



Reason R: Octet of each carbon is complete in



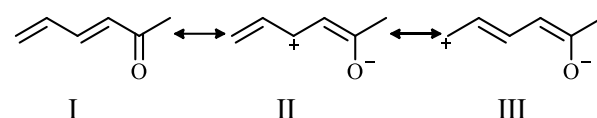
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

80. are related as:

- (1) position isomers
- (2) metamers
- (3) tautomers
- (4) same compounds

81. Relative stability order of given resonating structures is:



- (1) I > II > III
- (2) II > I > III
- (3) III > I > II
- (4) III > II > I

82. Minimum number of carbon required in alkanes to show optical isomerism is:

- (1) 4 (2) 6
(3) 7 (4) 1

83. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Acetate ion is more stable than phenoxide ion.

Reason R: Acetate ion has two equivalent resonating structures.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

84. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

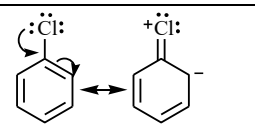
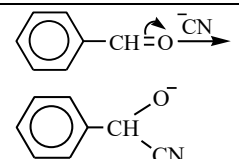
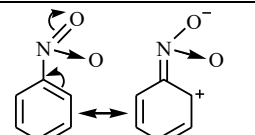
Assertion A: 2-Methyl pentane and 3-methyl pentane are position isomers of each other.

Reason R: Position of methyl group in the two compounds are different.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

85. Match the List-I with List-II.

List-I (Mechanism step)	List-II (Effect)
(A) 	(I) -R effect
(B) $\text{CH}_3-\text{CH}=\text{CH}_2 \xrightarrow{\text{H}^+}$ $\text{CH}_3-\text{CH}^+-\text{CH}_3$	(II) -E effect
(C) 	(III) +E effect
(D) 	(IV) +R effect

Choose the **correct** answer from the options given below;

- (1) A-IV; B-III, C-II; D-I
(2) A-IV; B-II, C-III; D-I
(3) A-I; B-II, C-III; D-IV
(4) A-I; B-III, C-II; D-IV

86. Number of structural isomerism of C_7H_{16} is:

- (1) 8
(2) 9
(3) 10
(4) 11

87. Most basic compound in gas phase among the following is:

- (1) CH_3NH_2
(2) $(\text{CH}_3)_2\text{NH}$
(3) $(\text{CH}_3)_3\text{N}$
(4) NH_3

88. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

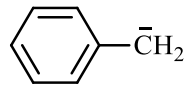
Assertion A: Ethanol and dimethyl ether are functional isomers of each other.

Reason R: Both have same molecular formula.

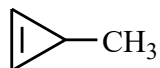
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

89. In which of the given species, carbanion is sp^3 hybridised?

- (1) $\text{CH}_3-\bar{\text{C}}\text{H}_2$
(2) $\text{CH}_2=\text{CH}-\bar{\text{C}}\text{H}_2$
(3) 
(4) All of these

90. Which of the following is **not** an isomer of butyne?

- (1) 
(2) 
(3) $\text{CH}_2=\text{C}=\text{CH}-\text{CH}_3$
(4) 

91. How many of the following statements regarding algae are **correct**?
- Algae are chlorophyll-bearing, simple, thalloid, autotrophic and largely aquatic organisms.
 - Some algae occur in association with fungi (lichen) and animals.
 - The form and size of algae is highly variable.
 - The algae reproduce only by vegetative and asexual methods.
 - Volvox* is a colonial alga.

Choose the **correct** option.

- Two only
 - Three only
 - Four only
 - Five
92. The earliest systems of classification which were based on gross superficial morphological characters such as habit, colour, number and shape of leaves, etc., are called:
- Natural classification systems.
 - Artificial classification systems.
 - Phylogenetic classification systems.
 - Cytotaxonomy.

93. Match **List-I** with **List-II**.

List-I (Product/Substance)		List-II (Obtained from)	
(A)	Peat	(I)	Brown algae
(B)	Algin	(II)	Red algae
(C)	Carrageen	(III)	Species of a moss
(D)	Fruits	(IV)	Angiosperms

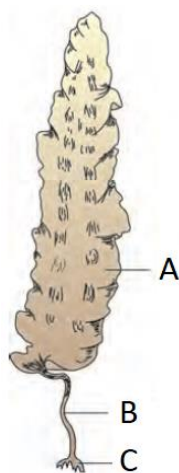
Choose the *most appropriate* answer from the options given below.

- A-III, B-I, C-II, D-IV
 - A-IV, B-II, C-I, D-III
 - A-I, B-III, C-IV, D-II
 - A-II, B-IV, C-III, D-I
94. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
- Assertion (A):** Bryophytes usually occur in dry localities.
- Reason (R):** Bryophytes can live in soil but are dependent on water for sexual reproduction.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- Both (A) and (R) are true and (R) is the correct explanation of (A).
 - Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (A) is true but (R) is false.
 - (A) is false but (R) is true.
95. The spread of living pteridophytes is limited and restricted to narrow geographical regions because:
- the sporophyte is dependent on the gametophyte.
 - they lack vascular tissues for water conduction.
 - their gametophytes require cool, damp, shady places to grow and need water for fertilisation.
 - they produce a large number of spores which fail to germinate in water.
96. Given below are two statements:
- Statement I:** In gymnosperms, the male and the female gametophytes have an independent free-living existence.
- Statement II:** The gymnosperms are heterosporous.
- In the light of the above statements, choose the **correct** answer from the options given below:
- Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
 - Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.
97. Arrange the following stages in the life cycle of a moss starting with the predominant phase in it.
- Zygote formation
 - Gamete fusion
 - Spore germination
 - Development of leafy stage
 - Spore formation in capsule
- Choose the **correct** option.
- C – D – B – A – E
 - C – D – A – B – E
 - E – C – D – B – A
 - E – D – C – A – B

98. Identify the organism shown in the given diagram and the **correct** function/characteristic of the given label.



- (1) *Porphyra*, B - Attachment to substratum
 (2) *Laminaria*, A - Photosynthesis
 (3) *Polysiphonia*, C - Leaf-like organ
 (4) *Fucus*, B - Connects frond and holdfast
99. Which of the following structures are **not** motile?
 (1) Asexual spores in most brown algae
 (2) Gametes in brown algae
 (3) Gametes in red algae
 (4) Asexual spores in green algae
100. Identify the **correctly** matched pair w.r.t stored food material.
 (1) Green algae - Floridean starch
 (2) Brown algae - Laminarin
 (3) Red algae - Cellulose
 (4) Green algae - Mannitol
101. Read the following statements about pteridophytes.
 A. The main plant body is a sporophyte.
 B. The sporophyte is differentiated into true root, stem and leaves.
 C. The leaves are large in *Selaginella*.
 D. Sporophytes bear sporangia that are subtended by sporophylls.
 E. In majority of the pteridophytes all the spores are of similar kinds.
 In the light of above statements choose the **correct** answer from the options given below:
 (1) A, B, D, E are correct
 (2) Only A, B, C are correct
 (3) Only B, C, D are correct
 (4) Only A, D, E are correct

102. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): In flowering plants, the pollen grains and ovules are developed in specialised structures called flowers.

Reason (R): The angiosperms are an exceptionally large group of plants occurring in wide range of habitats.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
 (3) (A) is true but (R) is false.
 (4) (A) is false but (R) is true.
103. The gemmae are;
 (1) sexual spores in bryophytes.
 (2) produced by all bryophytes.
 (3) green and multicellular buds.
 (4) produced in specialised structures called capsule.

104. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Psilopsida	(I)	<i>Equisetum</i>
(B)	Lycopsida	(II)	<i>Dryopteris</i>
(C)	Sphenopsida	(III)	<i>Psilotum</i>
(D)	Pteropsida	(IV)	<i>Selaginella</i>

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-IV, D-II
 (2) A-IV, B-II, C-III, D-I
 (3) A-I, B-III, C-II, D-IV
 (4) A-III, B-IV, C-I, D-II
105. Given below are two statements:
Statement I: Fusion of two gametes dissimilar in size, as in species of *Eudorina* is termed as oogamy.
Statement II: Fusion between one large, non-motile female gamete and a smaller, motile male gamete is termed as anisogamy.
 In the light of the above statements, choose the **correct** answer from the options given below:
 (1) Both Statement I and Statement II are correct.
 (2) Both Statement I and Statement II are incorrect.
 (3) Statement I is correct but Statement II is incorrect.
 (4) Statement I is incorrect but Statement II is correct.

106. At least a half of the total carbon dioxide fixation on earth is carried out by _____ through photosynthesis.
- algae.
 - bryophytes.
 - pteridophytes.
 - angiosperms.
107. In mosses;
- antheridia and archegonia are produced at the apex of leafy shoots.
 - a proper mechanism of spore dispersal is not present.
 - the sporophyte is unicellular.
 - the gametes are diploid.
108. Which of the following statements about gymnosperms is/are incorrect?
- The ovules are not enclosed by any ovary wall and remain exposed.
 - The giant redwood tree *Sequoia* is one of the tallest tree species.
 - Roots in *Pinus* have a fungal association in the form of mycorrhiza.
 - The stems are always unbranched.
 - They are called flowering plants.
- Choose the **correct** option.
- A and B
 - Only E
 - C and D
 - D and E
109. Identify the **incorrect** statement.
- Sporophylls form strobili in *Equisetum*.
 - In *Salvinia*, an event considered to be precursor to seed habit is observed.
 - Selaginella* does not possess vascular tissues.
 - The main plant body of fern possesses true stem, roots and leaves.
110. Read the given characteristics.
- Called amphibians of plant kingdom.
 - Include horsetails.
 - Main plant body is thalloid.
- Identify the plant groups in A, B and C and choose the **correct** option.
- A – Algae; B – Pteridophytes; C – Gymnosperms
 - A – Bryophytes; B – Pteridophytes; C – Algae
 - A – Algae; B – Pteridophytes; C – Angiosperms
 - A – Gymnosperms; B – Angiosperms; C – Bryophytes

111. Match **List-I** with **List-II**.

List-I		List-II	
(A)	<i>Chara</i>	(I)	Smallest plant that produces flowers
(B)	<i>Wolffia</i>	(II)	Thallus is dorsiventral and closely appressed to substratum
(C)	<i>Marchantia</i>	(III)	Possesses two types of shoots
(D)	<i>Ginkgo</i>	(IV)	Main pigments are chlorophyll <i>a</i> and <i>b</i>

Choose the *most appropriate* answer from the options given below.

- A-IV, B-II, C-I, D-III
 - A-IV, B-I, C-II, D-III
 - A-II, B-IV, C-I, D-III
 - A-I, B-II, C-III, D-IV
112. Choose the true statement about the plant shown below.



- It is a homosporous bryophyte.
 - The female gametophytes in these plants are retained on the parent sporophytes for variable periods.
 - The development of the zygotes into young embryos take place within the male gametophytes.
 - Protonema is the first stage of life cycle.
113. Identify the **incorrectly** matched pair.
- Main plant body is sporophytic - *Cycas*
 - Oogamous sexual reproduction - *Volvox*
 - Main plant body is gametophytic - *Polytrichum*
 - Isogamous sexual reproduction - *Fucus*
114. Given below are two statements:
- Statement I:** Natural classification systems were based on natural affinities among the organisms and considered both external and internal features.
- Statement II:** Phylogenetic classification systems are based on evolutionary relationships between the various organisms.
- In the light of the above statements, choose the **correct** answer from the options given below:
- Both Statement I and Statement II are correct.
 - Both Statement I and Statement II are incorrect.
 - Statement I is correct but Statement II is incorrect.
 - Statement I is incorrect but Statement II is correct.

115. In gymnosperms, the reduced gametophyte known as the pollen grain develops within the:
- (1) Megasporangium.
 - (2) Ovule.
 - (3) Microsporangium.
 - (4) Archegonium.

116. Consider the following statements about bryophytes:
- A. The main plant body is haploid.
 - B. The sporophyte is free-living and photosynthetic.
 - C. The male sex organ is called antheridium.
 - D. Zygotes undergo meiosis immediately to form haploid spores.
 - E. Bryophytes are divided into liverworts and mosses.

How many of the above statements are **correct**?

- (1) Two only
 - (2) Three only
 - (3) Four only
 - (4) Five
117. Choose the **odd** one out w.r.t conifers.
- (1) Stomata are sunken.
 - (2) Thick cuticle is present.
 - (3) Leaves have a large surface area.
 - (4) Leaves are needle-like.

118. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Chemotaxonomy uses the chemical constituents of the plant to resolve confusions.

Reason (R): In chemotaxonomy, number and codes are assigned to all the characters and the data are then processed.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

119. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Pyrenoids	(I)	Pteridophytes
(B)	Gemma cups	(II)	Liverworts
(C)	Prothallus	(III)	Gymnosperms
(D)	Naked seeds	(IV)	Green algae

Choose the *most appropriate* answer from the options given below.

- (1) A-III, B-I, C-II, D-IV
- (2) A-IV, B-II, C-I, D-III
- (3) A-III, B-II, C-I, D-IV
- (4) A-I, B-III, C-II, D-IV

120. In gymnosperms, the pollen tube discharges their contents;
- (1) inside the pollen sac.
 - (2) near the mouth of the archegonia.
 - (3) on the microsporangium.
 - (4) in water.

121. Read the given statements regarding the economic importance of algae.

- A. *Laminaria* and *Sargassum* are marine algae used as food.
- B. Algae increase the level of dissolved oxygen in their immediate environment.
- C. Agar is obtained from *Gelidium* and *Gracilaria*.
- D. *Chlorella* is a multicellular alga rich in proteins.
- E. Agar is used to grow microbes and in preparations of ice-creams and jellies.

The **incorrect** statement(s) is/are;

- (1) D only
- (2) A and E
- (3) B and C
- (4) A only

122. The _____ assumes that organisms belonging to the same taxa have a common ancestor.

- (1) classification system proposed by Linnaeus
- (2) natural classification system
- (3) phylogenetic classification system
- (4) classification system proposed by Bentham and Hooker

123. Given below are two statements:

Statement I: Mosses prevent soil erosion.

Statement II: The sporophyte in mosses is differentiated into a foot, seta and capsule.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both Statement I and Statement II are correct.
- (2) Both Statement I and Statement II are incorrect.
- (3) Statement I is correct but Statement II is incorrect.
- (4) Statement I is incorrect but Statement II is correct.

124. Identify the **incorrect** statement.

- (1) Kelps may reach a height of 100 metres.
- (2) *Eucalyptus* trees are tall (over 100 metres).
- (3) Kelps are freshwater organisms.
- (4) *Eucalyptus* is an angiosperm.

125. Read the following statements w.r.t Phaeophyceae.

- A. They are found primarily in marine habitats.
- B. They show great variation in size and form.
- C. They possess chlorophyll *a*, *b* and fucoxanthin.
- D. They vary in colour from olive green to various shades of brown depending upon the amount of the xanthophyll pigment, fucoxanthin.
- E. The vegetative cells lack cell wall.

Select the option having all **correct** statements.

- (1) A, B, C
- (2) A, B, D
- (3) B, C, D
- (4) C, D, E

126. The female gametophyte in *Cedrus*;

- (1) is unicellular.
- (2) bears only one archegonium.
- (3) is retained within megasporangium.
- (4) is absent.

127. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Angiosperms	(I)	Divided into 3 classes
(B)	Algae	(II)	Divided into 2 classes
(C)	Pteridophytes	(III)	Divided into 4 classes

Choose the *most appropriate* answer from the options given below.

- (1) A-II, B-I, C-III
- (2) A-I, B-II, C-III
- (3) A-II, B-III, C-I
- (4) A-I, B-III, C-II

128. In contrast to bryophytes, the main plant body in pteridophytes is:

- (1) haploid.
- (2) triploid.
- (3) tetraploid.
- (4) diploid.

129. What is common between *Porphyra* and *Gracilaria*?

- (1) 2-8 apical flagella
- (2) The flagella are of equal size
- (3) Complex post-fertilisation changes
- (4) Sexual reproduction is anisogamous

130. The rhizoids in *Funaria*;

- (1) are not branched.
- (2) are multicellular.
- (3) bear capsule at apex.
- (4) are not found.

131. The natural classification system for flowering plants was given by:

- (1) Carolus Linnaeus
- (2) R. H. Whittaker
- (3) George Bentham and Joseph Dalton Hooker
- (4) Aristotle

132. The function of air bladder present in the thallus of *Fucus* is;

- (1) to provide buoyancy.
- (2) attachment to substratum.
- (3) transport of water.
- (4) storage of photosynthetic pigments.

133. The cell wall in members of Chlorophyceae is usually made of outer layer of ____ and inner layer of ____.

- (1) pectose, cellulose
- (2) cellulose, pectose
- (3) pectose, algin
- (4) algin, cellulose

134. Floridean starch is structurally similar to;

- (1) cellulose and starch.
- (2) amylopectin and glycogen.
- (3) laminarin and starch.
- (4) mannitol and cellulose.

135. In bryophytes, the flask-shaped female sex organ that produces a single egg is called:

- (1) antheridium.
- (2) oogonium.
- (3) sporangium.
- (4) archegonium.

136. A new experimental drug is administered that acts as a competitive antagonist to GnRH, binding to its receptors without activating them. Which of the following effects would be the most immediate and direct consequence in an adult male?

- (1) A rapid increase in testosterone levels due to a feedback loop disruption.
- (2) A significant decrease in the secretion of LH and FSH from the adenohypophysis.
- (3) Hypertrophy of the seminiferous tubules and interstitial cells.
- (4) A decrease in the release of oxytocin and vasopressin from the neurohypophysis.

137. The hypothalamic-hypophyseal portal system is a specialized vascular connection crucial for the transport of which of the following?

- (1) All pituitary hormones to their respective target organs throughout the body.
- (2) Releasing and inhibiting hormones from the hypothalamus to the anterior pituitary.
- (3) Neurotransmitters from hypothalamic neurons directly to the posterior pituitary.
- (4) Oxytocin and vasopressin from their site of synthesis to their site of release.

138. Which of the following hormones is **correctly** matched with its primary function?

- (1) Prolactin (PRL) – Stimulates the synthesis and secretion of steroid hormones called glucocorticoids
- (2) Growth Hormone (GH) – Stimulates the synthesis and secretion of hormones from the thyroid gland
- (3) Thyroid Stimulating Hormone (TSH) – Stimulates the thyroid gland to secrete thyroxine and triiodothyronine
- (4) Adrenocorticotrophic Hormone (ACTH) – Responsible for milk production and development of mammary glands

139. Which of the following statements **correctly** differentiate between neural and hormonal control systems?

- A. Neural coordination provides a rapid, point-to-point connection for quick responses.
- B. Hormonal coordination involves the release of chemicals into the blood for a slower, more sustained response.
- C. The "All-or-None" principle applies to hormonal action, meaning a minimal amount of hormone elicits a maximal response.
- D. Both systems are completely independent and do not influence each other.

- (1) A and B only
- (2) A, B and C only
- (3) B and D only
- (4) A, B, C and D

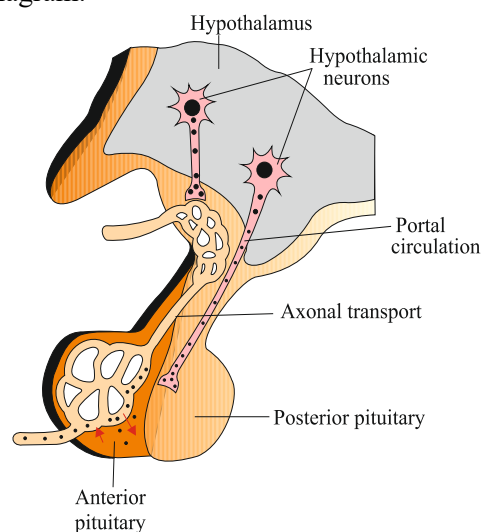
140. Match List-I with List-II.

List-I		List-II	
(A)	Sella tursica	(I)	Releases hormones synthesised in hypothalamic nuclei
(B)	Pars distalis	(II)	A bony depression in the sphenoid bone housing the pituitary
(C)	Pars intermedia	(III)	Region responsible for synthesizing the majority of pituitary hormones, including tropic hormones
(D)	Neurohypophysis	(IV)	Almost merged with the pars distalis in humans, secretes MSH

Choose the **correct** answer from the options given below:

- (1) A-II, B-III, C-IV, D-I
- (2) A-III, B-II, C-I, D-IV
- (3) A-II, B-IV, C-III, D-I
- (4) A-IV, B-III, C-I, D-II

141. Find out the **incorrect** labelling in the following diagram.



- (1) Axonal transport and Anterior pituitary.
- (2) Portal circulation and Posterior pituitary.
- (3) Anterior pituitary and Posterior pituitary.
- (4) Portal circulation and Axonal transport.

- 142.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In females, a mid-cycle surge of Luteinizing Hormone (LH) is the only trigger for oogenesis.
Reason R: Following ovulation, LH stimulates the transformation of the ruptured Graafian follicle into the corpus luteum and maintains it.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) A is true but R is false.
 - (2) A is false but R is true.
 - (3) Both A and R true and R is the correct explanation of A.
 - (4) Both A and R true and R is not the correct explanation of A.
- 143.** Consider the consequences of a lesion that completely severs the hypothalamic-hypophyseal portal system, leaving the rest of the hypothalamus and pituitary intact. Which of the following statements **correctly** describe the likely hormonal outcomes?
- A. Secretion of TSH and ACTH would decrease significantly.
 - B. Secretion of Growth Hormone would become unregulated and increase.
 - C. Secretion of ADH would remain largely unaffected.
- (1) A and C only
 - (2) A, B and C
 - (3) B only
 - (4) A and B only
- 144.** On the basis of the named structures, choose the option that **correctly** matches the physiological function of hormones:
- A = Hypothalamus
 B = Hypothalamic neuron
 C = Portal circulation
 D = Posterior pituitary
 E = Anterior pituitary
- (1) A – Only releases
 D – Release and synthesis
 E – Only synthesis
 - (2) A – Only release
 D – Only synthesis
 E – Synthesis and release
 - (3) A – Synthesis and release
 D – Only releases
 E – Synthesis and releasing
 - (4) A – Synthesis and release
 D – Only synthesis
 E – Only release
- 145.** Islets of Langerhans in a normal human pancreas comprise only:
- (1) 30–40% of pancreatic tissue
 - (2) 1–2% of pancreatic tissue
 - (3) 50% of pancreatic tissue
 - (4) 90% of pancreatic tissue
- 146.** Glucagon primarily targets hepatocytes (liver cells) to exert its hyperglycemic effect. Which of the following metabolic pathways is stimulated by glucagon?
- (1) Glycolysis and glycogenesis
 - (2) Cellular uptake of glucose and amino acids
 - (3) Lipogenesis and protein synthesis
 - (4) Glycogenolysis and gluconeogenesis
- 147.** An individual with a condition known as diabetes insipidus shows symptoms of polyuria (excessive urination) and polydipsia (excessive thirst). This can be caused by either a deficiency of ADH or the kidney's inability to respond to it. How could a physician differentiate between these two causes?
- (1) Measure the patient's blood glucose levels; high glucose indicates ADH deficiency.
 - (2) Check the patient's blood pressure; hypertension points towards kidney insensitivity.
 - (3) Administer a dose of synthetic ADH and monitor urine output.
 - (4) Analyze the blood for osmolality; low osmolality suggests ADH deficiency.
- 148.** Which of the following pair of hormones can raise blood glucose levels?
- (1) Aldosterone and Thyrocalcitonin
 - (2) Parathyroid Hormone and Prolactin
 - (3) Insulin and Somatostatin
 - (4) Cortisol and Adrenaline
- 149.** The composition of pancreatic juice is tailored to the composition of the chyme entering the duodenum, thanks to the differential effects of two key hormones. Which of the following statements **best** describe this coordinated regulation?
- (1) Secretin stimulates enzyme release, while CCK stimulates bicarbonate release.
 - (2) Both Secretin and CCK stimulate the release of an identical pancreatic juice.
 - (3) Acidic chyme stimulates Secretin for bicarbonate release, while fatty/protein-rich chyme stimulates CCK for enzyme release.
 - (4) Gastrin is the primary regulator, stimulating both bicarbonate and enzyme release from the pancreas.
- 150.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Thymosins are secreted from thyroid gland in old individual.
Reason R: Thymosins stimulates the production of antibodies.
 In the light of the above statements, choose the **correct** answer from the options given below:
- (1) Both A and R are true and R is the correct explanation of A.
 - (2) Both A and R are true but R is NOT the correct explanation of A.
 - (3) A is true but R is false.
 - (4) A is false but R is true.

151. Which of the following statements about the thymus gland is **correct**?
- (1) It is a large gland that remains active and grows throughout a person's life
 - (2) It is located in the abdominal cavity, just superior to the kidneys
 - (3) It is primarily responsible for the maturation of B-lymphocytes
 - (4) It secretes peptide hormones, which are crucial for the development of cell-mediated immunity

152. Growth factors secreted by non-endocrine tissues are vital for processes like wound healing. How does their mechanism of action typically differ from that of classical endocrine hormones like thyroxine?
- (1) Growth factors always act via intracellular receptors, while endocrine hormones use membrane-bound receptors.
 - (2) Growth factors are typically transported long distances via the bloodstream, whereas endocrine hormones act locally.
 - (3) Growth factors usually act locally on neighboring cells (paracrine) or on the same cell that secreted them (autocrine).
 - (4) Growth factors are nutrient-based chemicals, while endocrine hormones are non-nutrient chemicals.

153. Match List-I with List-II.

List-I (Hormone)		List-II (Role)	
(A)	Testosterone	(I)	Development of female secondary sex characters
(B)	Estrogen	(II)	Stimulates muscular growth
(C)	Progesterone	(III)	Stimulates Leydig cells
(D)	LH (in males)	(IV)	Supports pregnancy

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
 - (2) A-II, B-I, C-IV, D-III
 - (3) A-II, B-I, C-III, D-IV
 - (4) A-I, B-II, C-IV, D-III
154. Read the following statements and choose how many statements are **incorrect**.
- A. Testis performs dual functions as an endocrine gland as well as secondary sex organ.
 - B. High pitch voice is induced by testosterone.
 - C. The pituitary gland is attached to the thalamus with the help of stalk.
 - D. Androgenic steroids play a role in growth of axial hair, pubic hair and facial hair during puberty.
- (1) One
 - (2) Two
 - (3) Three
 - (4) Four

155. Given below are two statements:

Statement I: Thyroid hormones (T_3 and T_4) are essential for maintaining the basal metabolic rate and also influence water and electrolyte balance.

Statement II: The synthesis of T_3 , T_4 , and TCT are all critically dependent on the dietary intake of iodine.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
 - (2) Statement I is correct but Statement II is incorrect.
 - (3) Statement I is incorrect but Statement II is correct.
 - (4) Statement I and Statement II both are incorrect.
156. Prolonged hyperglycemia, as seen in untreated diabetes mellitus, can lead to the formation of harmful acidic compounds and their appearance in the urine. What are these compounds called, and what is the underlying metabolic shift that causes their formation?
- (1) Compounds: Uric acid; Shift: Breakdown of nucleic acids
 - (2) Compounds: Lactic acid; Shift: Anaerobic glycolysis in skeletal muscles
 - (3) Compounds: Ketone bodies; Shift: Excessive breakdown of fatty acids for energy
 - (4) Compounds: Amino acids; Shift: Excessive breakdown of proteins

157. A patient undergoes a total thyroidectomy (removal of the thyroid gland). Due to the close anatomical proximity, the parathyroid glands are accidentally removed as well. Which of the following statements describe the immediate and most critical consequences?

- A. The patient will experience a rapid drop in blood calcium levels (hypocalcemia).
 - B. Due to hypocalcemia, the patient is at high risk for tetany, characterized by skeletal muscle spasms and convulsions.
 - C. The patient will also suffer from a lack of thyrocalcitonin (TCT), leading to dangerously high blood calcium levels.
 - D. The metabolic rate will increase sharply due to the sudden removal of thyroid hormones.
- (1) A and B only
 - (2) A and D only
 - (3) B and C only
 - (4) A, B and C only

- 158.** A patient is diagnosed with a tumor of the adrenal medulla. Which of the following hormonal imbalances would be expected in this patient?
- (1) Overproduction of aldosterone, leading to hypertension and hypokalemia.
 - (2) Underproduction of cortisol, leading to hypoglycemia and weakness.
 - (3) Overproduction of catecholamines, leading to episodes of high blood pressure, palpitations, and sweating.
 - (4) Overproduction of androgenic steroids, leading to virilization in females.
- 159.** Which of the following **correctly** describes a dual function of Gastric Inhibitory Peptide (GIP)?
- (1) Stimulates gastric acid secretion and inhibits insulin release.
 - (2) Inhibits gastric motility and stimulates insulin release.
 - (3) Inhibits pancreatic enzyme secretion and increases bile flow.
 - (4) Stimulates glucagon secretion and gastric contractions.
- 160.** An elderly individual is noted to have a higher susceptibility to new infections and a reduced response to vaccinations. Simultaneously, under conditions of chronic stress, they exhibit elevated blood sugar and high blood pressure. Which of the following statements provide the most plausible physiological explanations for these observations?
- A. The age-related atrophy of the thymus gland has led to a decline in T-lymphocyte function, impairing the immune response.
 - B. Chronic stress has led to prolonged secretion of cortisol from the adrenal cortex, which has immunosuppressive and hyperglycemic effects.
 - C. Decreased secretion of aldosterone from the adrenal medulla is causing the blood pressure to rise.
 - D. The decline in immunity is due to the overproduction of catecholamines, which destroy lymphocytes.
- (1) A and B only
 - (2) C and D only
 - (3) A and D only
 - (4) B and C only

- 161.** Match **List-I** with **List-II**.

List-I (Layer)		List-II (Hormone Class)	
(A)	Zona glomerulosa	(I)	Glucocorticoids (e.g., Cortisol)
(B)	Zona fasciculata	(II)	Gonadocorticoids (e.g., Androgens)
(C)	Zona reticularis	(III)	Mineralocorticoids (e.g., Aldosterone)

Choose the **correct** answer from the options given below:

- (1) A-I, B-II, C-III
- (2) A-II, B-III, C-I
- (3) A-III, B-II, C-I
- (4) A-III, B-I, C-II

- 162.** The Renin-Angiotensin-Aldosterone System (RAAS) and Atrial Natriuretic Factor (ANF) constitute a crucial hormonal check-and-balance mechanism for regulating blood pressure. Which of the following statements **correctly** describe their antagonistic interaction?

- (1) ANF stimulates the release of renin, which then counteracts the effects of aldosterone.
- (2) Aldosterone and ANF both cause vasodilation but act on different blood vessels.
- (3) ANF is released in response to low blood pressure and acts to inhibit the RAAS pathway.
- (4) ANF is released in response to high blood pressure and acts to suppress the RAAS pathway.

- 163.** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

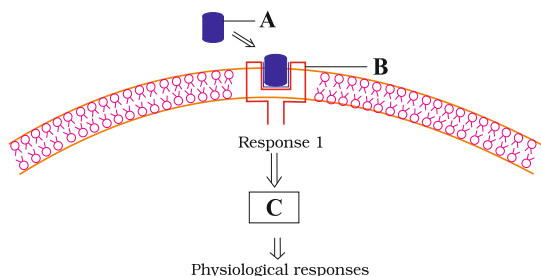
Assertion A: The hypothalamus is the basal part of the diencephalon of the forebrain.

Reason R: The hypothalamus controls and integrates activities of the autonomic nervous system and endocrine glands.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is not the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

164. Identify **A**, **B**, and **C** in the given below diagrammatic representation of the mechanism of hormone action.



- (1) A = Steroid Hormone; B = Hormone-receptor Complex; C = Protein
 - (2) A = Protein Hormone; B = Receptor; C = Cyclic AMP
 - (3) A = Steroid Hormone; B = Receptor; C = Second Messenger
 - (4) A = Protein Hormone; B = Cyclic AMP; C = Hormone-receptor Complex
165. Given below are two statements:
Statement I: Steroid hormones like aldosterone are lipid-soluble and bind to intrinsic receptors, mostly in the nucleus.
Statement II: Thymosin is a peptide hormone made of amino acids.
 In the light of the above statements, choose the most appropriate answer from the options given below:
- (1) Statement I and Statement II both are correct.
 - (2) Statement I is correct but Statement II is incorrect.
 - (3) Statement I is incorrect but Statement II is correct.
 - (4) Statement I and Statement II both are incorrect.
166. Considering the diverse functions of melatonin, which of the following statements is **incorrect**?
- (1) Melatonin levels are typically high during the day and low at night to regulate the diurnal rhythm.
 - (2) The pineal gland, which secretes melatonin, is located on the dorsal side of the forebrain.
 - (3) Besides the sleep-wake cycle, melatonin also influences pigmentation and the body's defense capabilities.
 - (4) The synthesis of melatonin is influenced by light exposure, which is why artificial light at night can disrupt sleep patterns.
167. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: As the nerve fibres do not innervate all cells of the body, and the cellular functions need to be continuously regulated, a special kind of coordination and integration has to be provided.
Reason R: The neural system provides chemical coordination and integration through hormones secreted directly into the blood.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
 - (2) Both A and R are true but R is NOT the correct explanation of A.
 - (3) A is true but R is false.
 - (4) A is false but R is true.
168. An athlete training at high altitudes for several weeks will experience a physiological adaptation to the lower atmospheric oxygen pressure. This adaptation primarily involves which of the following hormonal mechanisms?
- (1) Increased secretion of ANF from the heart to improve blood flow.
 - (2) Increased secretion of erythropoietin from the kidneys to enhance the blood's oxygen-carrying capacity.
 - (3) Decreased secretion of thyroid hormones to lower the body's overall metabolic oxygen demand.
 - (4) Increased secretion of aldosterone from the adrenal cortex to increase blood volume.
169. The regulation of gastric acid secretion is a complex interplay of stimulation and inhibition. Which pair of hormones has antagonistic effects on the secretion of hydrochloric acid (HCl) from parietal cells?
- (1) GIP and Secretin
 - (2) CCK and GIP
 - (3) Gastrin and Somatostatin
 - (4) Secretin and CCK
170. Which of the following statements **correctly** identifies the chemical nature and receptor location for the hormones glucagon and cortisol?
- (1) Glucagon is a peptide that binds to intracellular receptors, while cortisol is a steroid that binds to membrane receptors.
 - (2) Both are steroids; glucagon binds to membrane receptors and cortisol binds to intracellular receptors.
 - (3) Glucagon is a peptide that binds to membrane-bound receptors, while cortisol is a steroid that binds to intracellular receptors.
 - (4) Both are peptides; glucagon binds to intracellular receptors and cortisol binds to membrane receptors.

171. Given below are two statements:

Statement I: Adrenal cortex is the inner region of the adrenal gland.

Statement II: Endocrine glands are called ductless gland.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

172. Match List-I with List-II.

List-I (Type of hormone)		List-II (Example)	
(A)	Peptide hormone	(I)	Adrenaline
(B)	Steroid hormone	(II)	Insulin
(C)	Iodothyronine	(III)	Estradiol
(D)	Amino-acid derivative	(IV)	Thyroxine

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-IV, D-II
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-IV, C-I, D-II
- (4) A-II, B-III, C-I, D-IV

173. The modern definition of a hormone has expanded beyond the classical view. Which of the following are included under the current definition?

- A. Secretions from organized endocrine glands like the pituitary.
 - B. Chemicals released from ductless glands only.
 - C. Intercellular messengers produced by non-endocrine tissues that act locally.
 - D. Nutrient chemicals like glucose that act as signaling molecules.
- (1) A and B only
 - (2) A and C only
 - (3) B and D only
 - (4) A, C and D only

174. From the list of hormones given below, identify how many are synthesized and secreted by the pars distalis.

TSH, LH, FSH, ACTH, GH, Prolactin, Oxytocin

- (1) Two
- (2) Six
- (3) Four
- (4) Five

175. Match List-I with List-II.

List-I (Disorder)		List-II (Characteristic)	
(A)	Graves' disease	(I)	Hypersecretion of thyroid hormones, leading to increased BMR
(B)	Cretinism	(II)	Enlargement of the thyroid gland, protrusion of the eyeballs, increased basal metabolic rate, and weight loss
(C)	Hyperthyroidism	(III)	A condition characterized by decreased secretion of thyroid hormones in an adult
(D)	Hypothyroidism	(IV)	Mental retardation, low intelligence quotient, abnormal skin, deaf-mutism

Choose the **correct** answer from the options given below:

- (1) A-II, B-IV, C-I, D-III
- (2) A-II, B-IV, C-III, D-I
- (3) A-IV, B-I, C-II, D-III
- (4) A-IV, B-II, C-I, D-III

176. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The thyroid gland is composed of follicles and stromal tissues.

Reason R: Each thyroid follicle is composed of follicular cells enclosing a cavity, and these cells synthesize the hormone thyrocalcitonin (TCT).

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
- (2) Both A and R are true but R is NOT the correct explanation of A.
- (3) A is true but R is false.
- (4) A is false but R is true.

177. Insulin therapy is life-saving for patients with Type 1 diabetes mellitus, but overuse or incorrect dosage can lead to a dangerous condition. What is the most immediate and life-threatening risk associated with an overdose of insulin?

- (1) Severe hyperglycemia.
- (2) A sudden spike solely in blood pressure leading to stroke.
- (3) Profound hypoglycemia leading to a diabetic coma.
- (4) Rapid weight gain and suppression of the immune system.

178. Which of the following physiological processes is most critically impaired in a developing foetus due to maternal hypothyroidism, leading to the most severe and irreversible symptoms of cretinism?

- (1) Maturation and differentiation of central nervous system neurons.
- (2) Formation of red blood cells (erythropoiesis).
- (3) Regulation of the basal metabolic rate.
- (4) Maintenance of water and electrolyte balance.

179. The pituitary gland is called the “Master Gland” of the endocrine system. However, it is itself under the control of another gland. Which of the following **best** explains this relationship?

- (1) The pituitary gland controls the hypothalamus through releasing hormones.
- (2) The hypothalamus controls the pituitary gland through releasing and inhibiting hormones.
- (3) The pituitary gland functions completely independently of the hypothalamus.
- (4) The thyroid gland regulates both the hypothalamus and pituitary gland.

■■■

180. Match **List-I** with **List-II**.

List-I (Disease)		List-II (Cause)	
(A)	Gigantism	(I)	Hypersecretion of GH in adults mainly
(B)	Acromegaly	(II)	Hypersecretion of GH in children
(C)	Diabetes Insipidus	(III)	Hyposecretion of adrenal cortex hormones
(D)	Addison's Disease	(IV)	Deficiency of ADH

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-II, B-I, C-IV, D-III
- (3) A-II, B-I, C-III, D-IV
- (4) A-III, B-II, C-I, D-IV



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>

SPACE FOR ROUGH WORK

Test Analysis Sheet

- Student Name :
- PW Registration No. :
- Batch Code :
- Test Name/Type :

Max. Marks:
Marks Obtained:
Percentage:
Date of Exam :

1. Subject-wise Performance

Subject	Attempt Order	Time taken (min)	Marks Expected	No. of Attempted Q's	No. of Wrong Q's	Accuracy (%)
Physics						
Chemistry						
Botany						
Zoology						

- Was there a rush at the end? (Yes/No): _____
- Did you revise after solving? (Yes/No): _____

2. Wrong Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Calculation mistake					
Silly mistake					
Marked incorrect option, however solved correctly					
Conceptual Error					
Others					

3. Unattempted Questions Analysis

Reason	Physics	Chemistry	Botany	Zoology	Remark
Forgot the concept / Formula					
Solved correctly but didn't have confidence to mark it on OMR					
Missed the class /Didn't revise					
Got confused with the concept					
Others					

4. Weaknesses Identified

(List topics/chapters where you lost marks)

- Physics :
- Chemistry :
- Botany :
- Zoology :

5. Action Plan Before Next Test

1. Types of Mistakes to Avoid:
2. Extra Practice Needed in:
3. Strategy Changes: (e.g., attempt Chemistry first, manage time better):
4. What would you like to improve emotionally/mentally for the next test? :

Student Signature: _____